RDTII-BENCH: AN AUDITABLE HARNESS FOR MULTILINGUAL LEGAL EVIDENCE EXTRACTION

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ABSTRACT

Automated extraction of regulatory evidence from primary law is increasingly deployed across jurisdictions, but it is almost never evaluated across the scripts those jurisdictions actually publish in. We present a harness that treats a legal-evidence pipeline the way an experiment is treated: every published number is carried from a recorded run rather than typed, and every stage refuses rather than guesses. We instantiate it on the RDTII 2.1 regulatory framework over six Asia-Pacific economies publishing in three writing systems, and use it to ablate a single design decision — how text is tokenised for lexical retrieval — against the reference answers of the framework's own expert panel. Under the monolingual tokeniser the indicator vocabulary reaches 0.000 of China's cited instruments and 0.083 of Mongolia's; making tokenisation script-aware leaves China at 0.000 and Mongolia at 0.083, and only adding native statutory vocabulary raises them to 0.395 and 0.250. Both factors are necessary and neither is sufficient, and in three of the four configurations the failure is silent: the pipeline completes, raises nothing, and reports an absence of law that is indistinguishable from a genuine one. Widening the tokeniser is not free for the languages that already worked: with the ASCII path byte-identical, Australia's recall@10 moves by -0.105 and Malaysia's by +0.040, because the two share one index and therefore one set of collection statistics. Reachability is a property of the vocabulary and the tokeniser rather than of the corpus sample: over 5 seeded draws of the distractor pool its spread is 0.0000, while the rank-sensitive MRR over the same draws spreads 0.0005. The harness, not the pipeline, is the contribution: its methodology is domain-general, and we describe what porting it to another body of law requires.

1 INTRODUCTION

Regulatory indices are built by hand. An analyst is given an economy and a topic, searches official portals, reads statutes, and records which provision satisfies which indicator. The ESCAP Regional Digital Trade Integration Index alone spans eleven thematic pillars over forty-four sub-pillars, each requiring a coder to locate and justify primary law in the jurisdiction's own language (United Nations ESCAP, 2024). The work is slow, expensive and difficult to reproduce, and it is a natural target for automation: retrieve candidate provisions, ask a language model whether each satisfies a codified legal test, and export the accepted mappings with citations.

Systems of this shape are being built and sold. They are also, measurably, not yet trustworthy. In the first pre-registered evaluation of commercial AI legal-research tools, Magesh et al. (2024) found that products from LexisNexis and Thomson Reuters — retrieval-augmented, and marketed on their freedom from hallucination — fabricated or misattributed authority between 17% and 33% of the time. The response from the research community has mostly been better benchmarks of legal *reasoning* (Chalkidis et al., 2022; Guha et al., 2023) and better legal *retrieval* (Goebel et al., 2023). Those are necessary. They are also evaluations of the model, and the failures we describe here are not in the model.

Three failures recur in systems of this kind, and they compound:

- **The model authors the evidence.** A generated citation, article number or quotation cannot be checked against anything. Once a model is permitted to write the text it is also citing, the output is unfalsifiable — which is precisely the regime in which the hallucination rates above were measured.

- **The paper drifts from the run.** Numbers are copied into prose by hand, the experiment is re-run, and the prose is not. Surveys of machine-learning reproducibility find that no paper in a sample of four hundred documented everything needed to reproduce it, and that provenance over the artifacts of a pipeline — data, code, environment, seeds — is the missing discipline rather than an optional extra (Samuel et al., 2020). The abstract and the table disagree, and nobody notices because nothing checks.
- **The evaluation is monolingual.** The system is developed against English-language jurisdictions and shipped to others. Multilingual retrieval resources exist (Zhang et al., 2021; 2022), but a system evaluated per language, on a corpus already in that language, never encounters the condition that actually breaks: one index, several scripts, one tokenisation rule. The failure that follows is not a crash; it is a quiet zero.

This paper is about a harness that closes all three, and about what the harness found when it was pointed at its own pipeline.

1.1 Contributions

1. **An auditable harness for legal evidence extraction.** One invariant — every number in a published artifact is carried from a recorded run, never typed by a human and never written by a model — enforced by a checker that walks the paper source and fails the build on any value it cannot ground. To our knowledge this is the first application of run-level provenance enforcement to a legal-evidence pipeline, and it is the same rule the pipeline already applies to legal text, lifted one level up (Section 2).
2. **A multi-region, multi-script evaluation.** Six economies, three writing systems, nine indicators, graded against the reference answers recorded by the framework's own expert panel (Section 3).
3. **A measured account of a silent multilingual failure.** We show that a monolingual tokenisation assumption removes an entire economy from retrieval without raising anything, that fixing tokenisation alone does not recover it, and that the fix is not free for the languages that already worked (Section 4).
4. **A portable methodology.** The harness couples to a project through a single manifest; we set out what a different body of law has to supply (Section 5).

2 The harness

2.1 The invariant

The pipeline under study already runs on an anti-hallucination rule: legal text, citations and URLs are *carried from extraction, never generated*. The model grades evidence; it never authors it. A cited snippet must be verifiably present in the source document before the mapping is trusted, which makes every mapping checkable against a document rather than against the model's confidence in itself.

The harness applies exactly that rule one level up:

> Every number, table and figure in a published artifact is carried from a recorded run, never typed by a human and never written by a model.

Same function, different domain. The pipeline grounds legal snippets against source law; the harness grounds paper numbers against run records. No language model runs anywhere in the path that produces a number, a table or a claim — the harness is a deterministic toolchain, and models run only *inside* the experiments it schedules. This is what makes the reproducibility statement in this paper checkable rather than asserted.

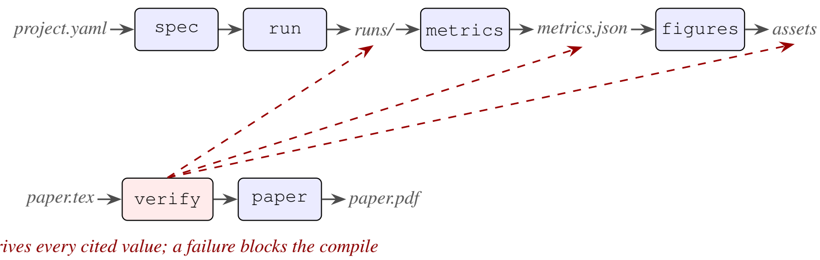

Figure 1: The harness. Solid arrows are artifacts flowing forward. Dashed arrows are `verify` walking backward from the prose to the records — resolving each cited value through `metrics.json` to the runs that produced it, and checking that every generated asset still matches. Prose is the only hand-written input, and it is the one input that is checked. No language model appears anywhere on this diagram: models run *inside* the experiments `run` schedules, never in the path that produces a number.

2.2 Stages

The harness is a set of commands with file inputs and file outputs, orchestrated by a makefile rather than a scheduler, so that any stage can be re-run alone and a broken stage never blocks the others. A project declares itself through one manifest; the harness never vendors project code, it imports the entry points the manifest names.

- `spec` validates the manifest and resolves entry points.
- `run` executes the experiment grid, one record per point. A record's filename *is* the hash of its task, its full configuration and its input data, so resume is correct rather than merely convenient: a record produced under a different configuration cannot land on the same name, and so cannot be silently reused.
- `metrics` collapses records into cells. The seed axes are aggregated away here and nowhere else, so by the time a number is visible to a figure or a claim it already carries its sample size, standard deviation, extremes and spread. A single lucky seed cannot be published as a fact.
- `figures` generates every asset the paper includes, deterministically.
- `verify` walks the paper source, resolves every cited value back through the metrics to the records behind it, and exits non-zero on anything it cannot ground.
- `paper` compiles the document, but runs `verify` first and refuses to produce a PDF if any number in the source is unbacked.

2.3 What refusal means

Three decisions are worth defending, because each is a place where the obvious implementation would have been to guess.

An ambiguous selector raises. A claim names a number by naming a condition. A condition matching two cells is an error, not a pick-the-first: a published number must not depend on dictionary ordering.

One seed measures nothing about variability. Where a cell has a single run, the standard deviation is reported as undefined rather than as zero. Zero would read as *we measured stability and found it*.

A failed run is recorded as a failure. It is never scored zero and never averaged in as though it were a measurement. A cell in which every run failed keeps its place in the output with no metrics attached, and any claim that cites it raises rather than quietly returning zero.

The same instinct governs the pipeline. An indicator with no supporting provision emits an explicit *no evidence found* row rather than a blank, and instruments that are draft or repealed are dropped from the output rather than flagged in a notes column — a warning nobody reads is not a control.

3 EXPERIMENTAL SETUP

3.1 TASK

RDTII 2.1 codifies nine indicators across two data-policy pillars — cross-border data movement, and domestic data protection. Each indicator carries a legal test: the operative rule a provision must enact to satisfy it. The retrieval question is prior to and strictly harder than the grading question, because it bounds it: a provision that never reaches the shortlist cannot be graded, so it can never be answered. We therefore measure the retrieval floor.

3.2 CORPUS AND GOLD

The evaluation corpus is built from two files, both checked into the repository, so that a run needs no network, no credentials and no language model, and can be re-derived by anyone who clones it.

The reference set is the expert panel's own record: for each economy and indicator, the instrument the panel accepted as evidence. It covers six economies — Singapore, Australia, Malaysia, India, China and Mongolia — and yields 49 gold pairs for India, 38 for China and 12 for Mongolia, among others.

A document is the title as its own portal serves it. This matters more than it sounds. The panel worked in English and recorded Chinese instruments bilingually, and Mongolian instruments under English names only. A crawler on a Chinese portal sees only the Chinese title; a crawler on the Mongolian portal sees only Mongolian. Indexing the panel's English glosses would measure the panel's spreadsheet rather than the corpus a system actually meets, and would conceal the failure this experiment exists to expose. We therefore keep the native title wherever the record carries one. For Mongolia, where it does not, we recover it by matching the numeric instrument identifier inside the source URL against the portal's own catalogue — a language-independent join, and the same one the project adopted after discovering that scoring Mongolia at zero had been a measurement error rather than a result. This recovers 5 of Mongolia's instruments into their own script; 29 of China's are native by construction. Unresolved instruments keep their English name and are counted, so the extent to which the Mongolian arm is genuinely a Cyrillic task is a reported number rather than an assumption.

Distractors are drawn from a harvested catalogue of Mongolian legal titles, which is exactly what a distractor pool should be: real legal titles in a real script, with no provision text, no indicator and no mapping. Each cell indexes the gold instruments of all six economies together with a seeded sample of that pool, giving 2113 documents. One pooled index rather than six is deliberate: it is the setting the claim is about, since a harness serving several regions holds them in one corpus, and the question is whether indexing Mongolia alongside Singapore degrades either.

3.3 CONDITIONS

The grid crosses economy with two binary treatments, repeated over 5 seeded draws of the distractor pool.

Tokenisation is either *monolingual* — the original rule, which matches runs of ASCII letters and digits — or *script-aware*, which routes spaced scripts to word tokens and no-space scripts to character bigrams. The script-aware tokeniser is imported from the shipped system rather than reimplemented: a benchmark that measures a copy of the component under test measures the copy. The monolingual rule is reproduced as the control arm, and the shipped implementation holds its ASCII path byte-identical to it so that previously swept retrieval parameters continue to hold.

Vocabulary is either the indicators' English query terms alone, or those terms plus native statutory phrases in the economy's own language. Native terms are added, never substituted, so that a wrong guess can only fail to match and can never displace a term that was working.

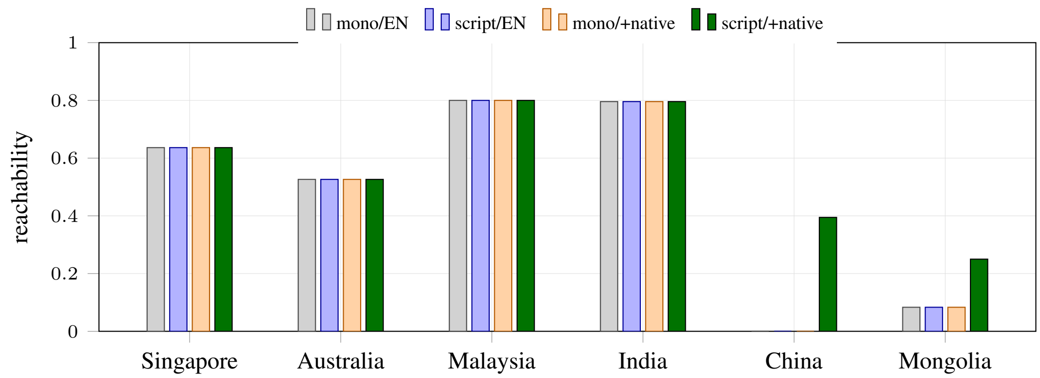

Figure 2: Reachability by economy under all four conditions. The four Latin-script economies are flat across every bar — the treatments are no-ops where there is nothing to translate and nothing the monolingual rule cannot read. China and Mongolia move only on the fourth bar, where both treatments are applied together. Every value plotted here is a `\lnum` macro resolved from the run records at compile time.

3.4 Metrics

- **Reachability** — the share of an economy's gold pairs whose instrument receives a non-zero lexical score. This is the silent-failure metric. A zero here is not a ranking that went badly; it is a document the query cannot see at all.
- **recall@10** — the share whose instrument reaches the top ten of the pooled index.
- **MRR** — mean reciprocal rank over the full ranking, zero when unreachable. Ties are broken pessimistically: an instrument tied with fifty others ranks behind all of them.

4 Results

4.1 An economy can vanish without an error

Table 1: Reachability by economy and condition. Latin-script economies are unaffected by either treatment, by construction. The two non-Latin economies are unreachable under three of four conditions — and no configuration raises, logs or fails.

		English terms		+ native terms	
Economy	Script	Monolingual	Script-aware	Monolingual	Script-aware
Singapore	Latin	0.6364	0.6364	0.6364	0.6364
Australia	Latin	0.5263	0.5263	0.5263	0.5263
Malaysia	Latin	0.8000	0.8000	0.8000	0.8000
India	Latin	0.7959	0.7959	0.7959	0.7959
China	Han	0.0000	0.0000	0.0000	**0.3947**
Mongolia	Cyrillic	0.0833	0.0833	0.0833	**0.2500**

Under the monolingual tokeniser the indicator vocabulary reaches 0.000 of China's cited instruments and 0.083 of Mongolia's; making tokenisation script-aware leaves China at 0.000 and Mongolia at 0.083, and only adding native statutory vocabulary raises them to 0.395 and 0.250.

The structure of Table 1 and Figure 2 is the finding. Script-aware tokenisation is necessary and not sufficient: China stays at 0.0000 once its text is tokenisable, because an English query and a Chinese statute share no tokens whatever the tokeniser does. Native vocabulary is likewise necessary and not sufficient: with the monolingual tokeniser, adding Chinese query phrases leaves China at 0.0000, because those phrases tokenise to nothing on the way in. Only the two together recover anything.

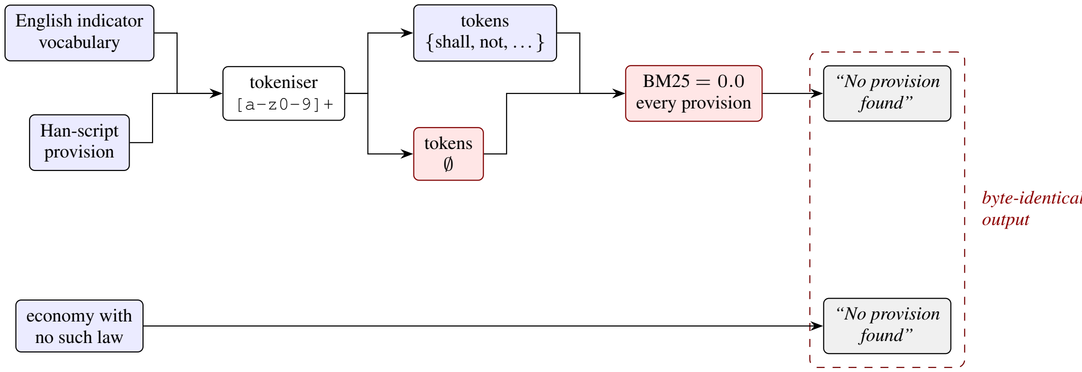

Figure 3: Why the failure survives review. A tokenisation mismatch and a genuine regulatory absence produce the same output, and an absence is a legitimate and common answer in a regulatory index. Nothing raises, nothing logs, and the exit code is zero on both paths.

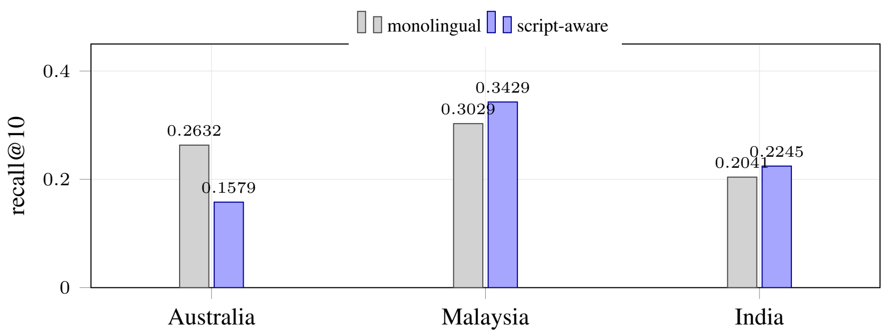

Figure 4: The cost of widening the tokeniser, on three economies whose own tokenisation did not change by a single byte. Admitting Cyrillic titles to the index alters the document-frequency and length-normalisation statistics that BM25 shares across the collection, and the ranking of Latin-script documents moves with them — down for Australia, up for Malaysia and India.

The operational point is not the magnitude but the mode. Lexical scoring contributes the majority of the hybrid retrieval score in the deployed system, so an economy in this state contributes a uniform zero to ranking. No exception is raised, no log line is emitted, and the pipeline terminates normally, reporting that no provision was found. That output is indistinguishable from the output for an economy that genuinely has no such law — which is a legitimate and common answer in a regulatory index, and is precisely why the failure survives review.

4.2 Widening the tokeniser is not free

Widening the tokeniser is not free for the languages that already worked: with the ASCII path byte-identical, Australia's recall@10 moves by -0.105 and Malaysia's by +0.040, because the two share one index and therefore one set of collection statistics.

Under the monolingual rule the Mongolian distractors tokenise to nothing: they occupy the index but can never score, and are effectively absent. Making the tokeniser script-aware admits them, which changes the document-frequency and length-normalisation statistics that BM25 shares across the whole collection (Robertson & Zaragoza, 2009). Australia's recall@10 falls from 0.2632 to 0.1579 although not one byte of Australian tokenisation changed; Malaysia rises from 0.3029 to 0.3429 and India from 0.2041 to 0.2245.

We report this because it is the kind of coupling that multi-region systems acquire silently. Reachability, being a per-document property, is unchanged for every Latin economy; only the rank-sensitive metrics move. A team measuring only reachability would have concluded the change was free, and a team measuring only recall on one economy would have concluded it was harmful.

4.3 THE HEADLINE METRIC IS STABLE

Reachability is a property of the vocabulary and the tokeniser rather than of the corpus sample: over 5 seeded draws of the distractor pool its spread is 0.0000, while the rank-sensitive MRR over the same draws spreads 0.0005.

Reachability does not depend on which distractors were drawn, which is what one should expect of a property of the query vocabulary and the tokeniser, and is worth confirming rather than assuming. The rank-sensitive metrics do move with the draw, which is why the seed is an axis of the grid rather than a fixed constant, and why every cell in this paper carries its spread.

5 EXTENDING TO OTHER DOMAINS OF LAW

Nothing in the harness is specific to trade regulation. A project couples to it through one manifest, and the manifest is the only coupling. Porting to a different body of law — competition, environmental, labour, tax — requires four things, and none of them are harness changes:

1. **A codified test per indicator.** The operative rule a provision must enact, written so that a disagreement between two graders is a disagreement about law rather than about wording. This is the expensive part, and it is legal work rather than engineering work.
2. **A reference set.** Instrument-level answers from a source that is authoritative in the domain. Ours is the framework's own expert panel; a competition-law port might use a regulator's decisional practice.
3. **An entry point.** Any callable that takes a configuration and returns metric names to numbers. The harness imports it; it never vendors the code.
4. **Native vocabulary per language.** The result above says this is not optional in any multilingual jurisdiction, and that it cannot be substituted for by a better tokeniser.

Two design properties do the portability work. Identity is content-addressed, so correctness of resume does not depend on remembering what was run. And the experiment grid is declared rather than coded, so adding a jurisdiction, a script or an ablation is a manifest edit.

We would flag one boundary honestly. The harness guarantees that a published number traces to a recorded run; it does not guarantee that the run measured anything worth knowing. Choosing indicators, building the reference set and deciding what counts as a correct mapping remain domain expertise. What the harness removes is the class of error where the number in the abstract stopped matching the experiment behind it.

6 RELATED WORK

Legal NLP benchmarks. LexGLUE assembles seven English legal understanding tasks (Chalkidis et al., 2022); LegalBench extends the idea to 162 tasks of legal *reasoning*, built with practising lawyers (Guha et al., 2023); COLIEE has run statute and case-law retrieval and entailment tasks for over a decade (Goebel et al., 2023). All three evaluate a model against a corpus that is given. Our object is the layer beneath: whether the pipeline that assembles the corpus can see the law at all. A benchmark whose corpus is handed to it cannot expose a discovery or tokenisation failure, because it has already been paid for.

Reliability of legal retrieval systems. Magesh et al. (2024) is the closest work in spirit — an audit of deployed systems rather than of models, finding 17–33% hallucination in tools sold as hallucination-free. The difference is the failure mode. Their subject is a system that answers when it should not; ours is a system that *declines* to answer when it should have, and does so in a form indistinguishable from a correct decline. A hallucination is visible to anyone who checks the citation. An unreachable jurisdiction is visible to nobody, because there is no citation to check.

Multilingual retrieval. Mr. TyDi (Zhang et al., 2021) and MIRACL (Zhang et al., 2022) established monolingual retrieval evaluation across typologically diverse languages, and MIRACL in particular covers eighteen. Both evaluate *per language*: queries and corpus share a language, and each language is scored on its own index. That design is right for measuring a retriever and wrong for our question, which only arises when several scripts share one index and one tokenisation rule, and where — as Section 4 shows — admitting one script changes the collection statistics that rank the others.

Closest to our mechanism is Ogundepo et al. (2022), who show that character n-gram tokenisation is a strong general substitute for language-specific tokenizers in retrieval. We take that result as given and ask a different question: not which tokeniser scores best, but what a deployed system reports when it has the wrong one. The answer is not a lower score. It is a uniform zero that the pipeline reports as an absence of law.

Attribution and grounding. Retrieval-augmented generation with citation grounding checks that a generated statement is supported by retrieved text, and the pipeline studied here applies exactly that control: a mapping is rejected unless its quoted snippet is verifiably present in the source document. Such controls assume retrieval succeeded. Where an economy is unreachable, a grounding check has nothing to object to — there is no unsupported claim, only an absence, and an absence is a valid and common answer in a regulatory index. Grounding is a guard on the answer; nothing in it guards the question.

Provenance and reproducibility. Experiment trackers, workflow engines and artifact stores record what was run, and provenance over a pipeline's artifacts is well understood as the missing discipline (Samuel et al., 2020). What none of that stack does is check what was *written*: nothing objects when a number is typed into a manuscript and the experiment is then re-run. Our contribution to that literature is narrow and, we think, load-bearing — extending provenance across the last hop, from the record to the sentence that cites it, and making the build fail rather than the reader guess.

7 LIMITATIONS

This measures the lexical floor, not end-to-end accuracy. The unit here is the instrument title as its portal serves it, not full provision text, and the metric is lexical reachability, not mapping F1. Provision-level extraction for China, India and Mongolia requires a built corpus that does not yet exist, and a live crawl is neither reproducible nor free. We report the layer we can measure reproducibly and decline to report the one we cannot; the harness makes that distinction visible rather than letting an unmeasured claim pass as a measured one.

Reachability is a floor, not a ceiling. A reachable instrument may still rank poorly, and a dense retrieval stage may recover documents that lexical scoring misses. The deployed system weights lexical scoring at the majority of its hybrid score, so a uniform zero there is materially damaging, but it is not the whole retrieval system.

Reference sets are not exhaustive. The panel recorded the instruments it accepted, not every instrument that would satisfy each test. A system may surface a legitimate instrument the panel did not record, which this evaluation scores as a miss. This biases all reported figures downward and biases them equally across conditions, so the comparisons hold even though the absolute levels understate.

The Mongolian native vocabulary is unvalidated. It is a seed list of statutory stems, and Mongolian is agglutinative, so a lexical index may hold an inflected form that the stem never matches. The Mongolian recovery reported here should be read as a lower bound from an unaudited vocabulary.

8 CONCLUSION

A legal-evidence pipeline that cannot be audited is a pipeline whose failures are invisible, and the multilingual failures are the most invisible of all, because they present as the same output a correct system produces on a jurisdiction with no such law. We built a harness that refuses to publish a number it cannot trace to a run, pointed it at our own retrieval layer, and found an economy that had been silently absent under a tokenisation rule that was correct for every jurisdiction it was developed against.

The finding is small and the discipline is not. Every number in this paper, including the ones in the abstract, was substituted from a record on disk at compile time, and the build fails if any of them stops resolving. That property is what we are proposing, more than the measurement it happens to carry.

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A CASE STUDIES

Three instruments, one per script, walked through the same code the benchmark measures. The aggregate result says an economy is unreachable; these show why, token by token. The Singaporean case is the control — Latin script throughout, so no condition changes anything.

The native titles are reproduced as images. This is not a stylistic choice: the TeX installation used for this build has no CJK package and no Cyrillic encoding, and transliteration is not available to us, because the pipeline's own rule is that statutory text is carried and never rewritten. What appears below is the statute's own text rather than a romanisation of it. Non-Latin tokens are counted rather than printed for the same reason, and counted separately from Latin ones so that a mixed-script query shows which half is doing the work.

A.1 CN — P6-I2 (HAN SCRIPT)

Instrument. Personal Information Protection Law of the PRC. The panel's own answer for local storage, and the article the shipped grading prompt uses as its worked example.

As the portal publishes it:
中华人民共和国个人信息保护法

Condition	Query tokens	Title tokens	Overlap
Monolingual, English terms	`local`, `storage`, `requirements`, `stored`, … (82)	*none*	**0**
Script-aware, English terms	`local`, `storage`, `requirements`, `stored`, … (82)	13 non-Latin	**0**
Monolingual, + native terms	`local`, `storage`, `requirements`, `stored`, … (82)	*none*	**0**
Script-aware, + native terms	`local`, `storage`, `requirements`, `stored`, … (82) + 33 non-Latin	13 non-Latin	**6**

A.2 MN — P7-I1 (CYRILLIC SCRIPT)

Instrument. Law on Personal Data Protection. Recovered into its own script by the portal-id join; the panel recorded it under an English name only.

As the portal publishes it:
ХҮНИЙ ХУВИЙН МЭДЭЭЛЭЛХАМГААЛАХ ТУХАЙ

Condition	Query tokens	Title tokens	Overlap
Monolingual, English terms	`comprehensive`, `legal`, `framework`, `for`, … (59)	*none*	**0**
Script-aware, English terms	`comprehensive`, `legal`, `framework`, `for`, … (59)	4 non-Latin	**0**
Monolingual, + native terms	`comprehensive`, `legal`, `framework`, `for`, … (59)	*none*	**0**
Script-aware, + native terms	`comprehensive`, `legal`, `framework`, `for`, … (59) + 10 non-Latin	4 non-Latin	**2**

A.3 SG — P7-I1 (LATIN SCRIPT)

Instrument. Personal Data Protection Act 2012. The control. Latin script throughout, so no condition changes anything.

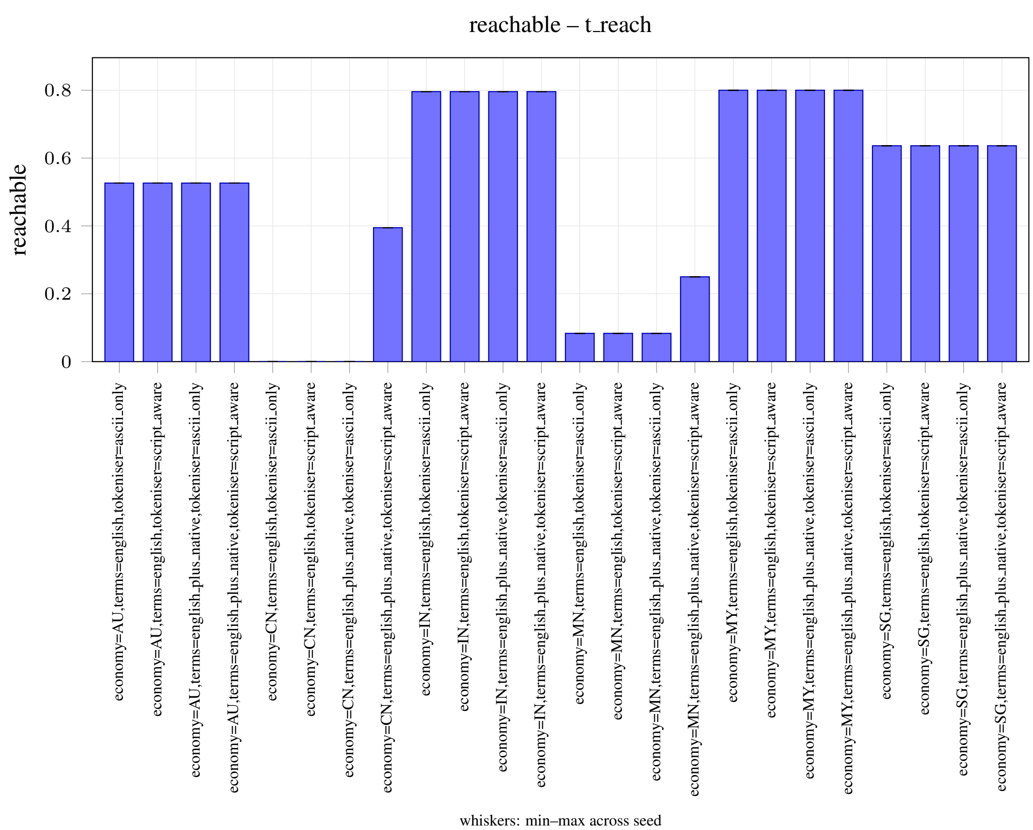
As the portal publishes it:
Personal Data Protection Act 2012

Condition	Query tokens	Title tokens	Overlap
Monolingual, English terms	`comprehensive, legal, framework, for,`... (59)	`personal, data, protection, act,` ... (5)	**4**
Script-aware, English terms	`comprehensive, legal, framework, for,`... (59)	`personal, data, protection, act,` ... (5)	**4**
Monolingual, + native terms	`comprehensive, legal, framework, for,`... (59)	`personal, data, protection, act,` ... (5)	**4**
Script-aware, + native terms	`comprehensive, legal, framework, for,`... (59)	`personal, data, protection, act,` ... (5)	**4**

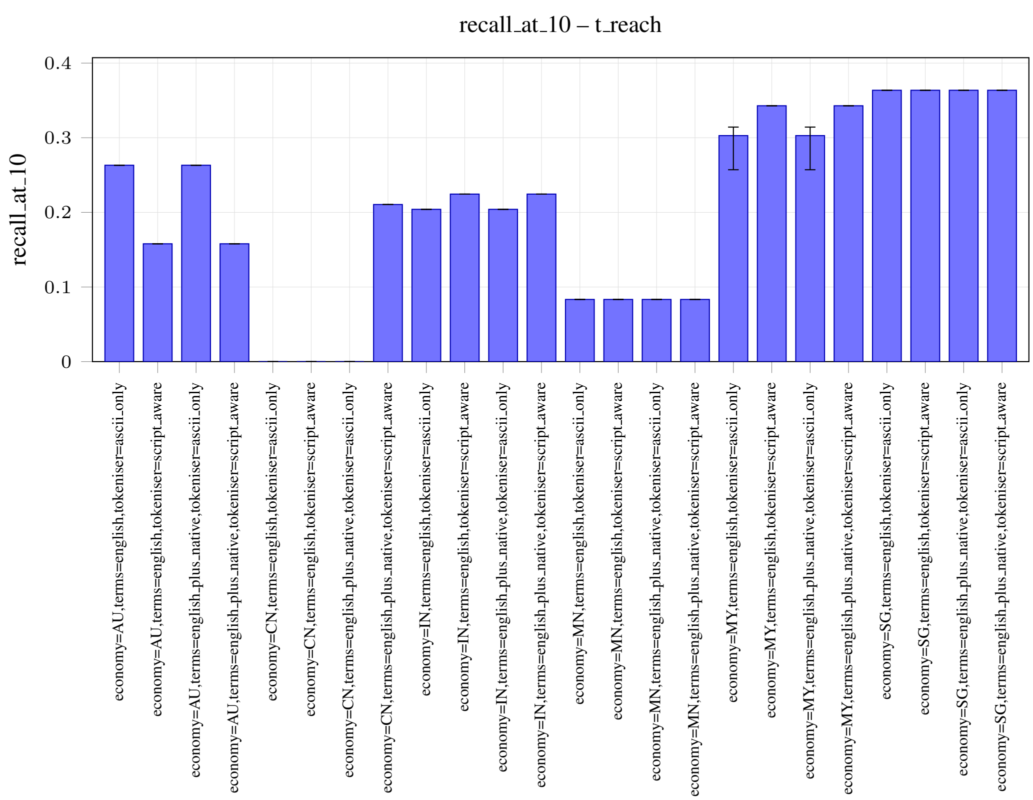
B Generated charts

Emitted by the `figures` stage as pgfplots, from the same metrics file as every table below. Bars are cell means and whiskers run min to max across the seed axes; a cell measured at one seed draws no whisker, because a whisker of zero would read as a measured spread of zero. Condition labels are the full cell configuration.

B.1 Reachability

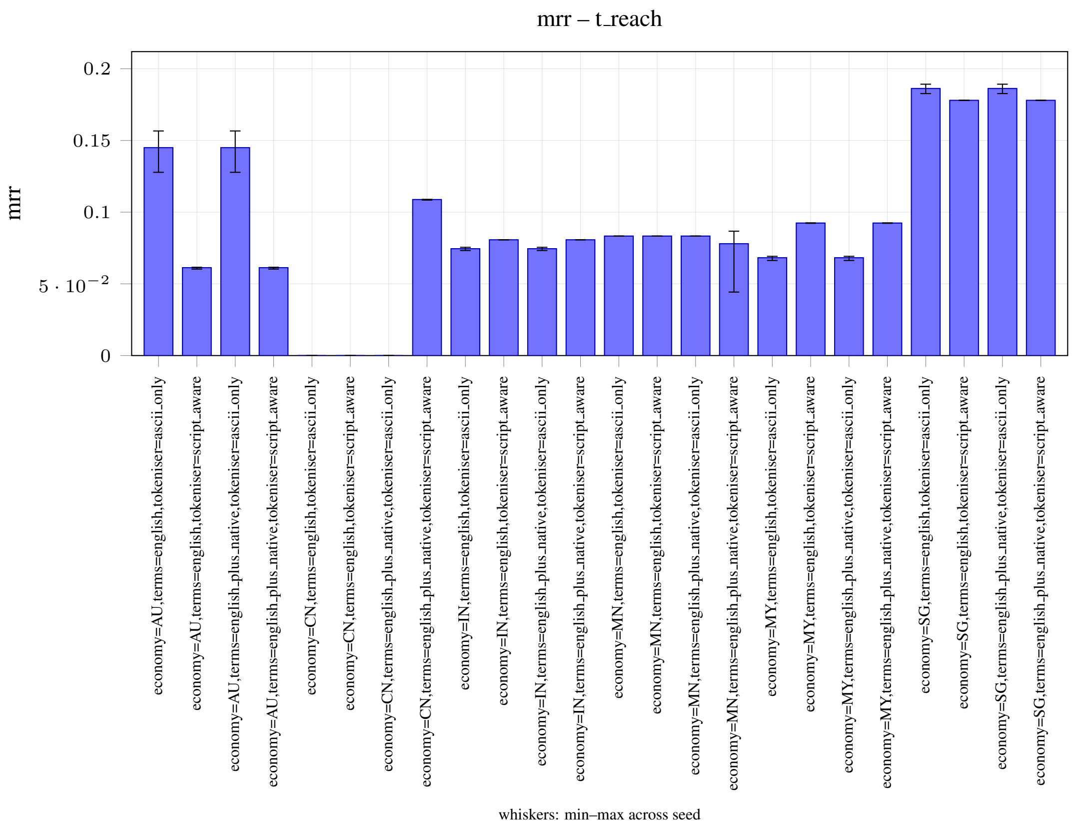


B.2 RECALL@10



whiskers: min–max across seed

B.3 MEAN RECIPROCAL RANK



C FULL CELL LISTING

The tables below are generated from the metrics file and are not hand-edited; the harness verifies at build time that each matches what the current records produce.

C.1 REACHABILITY

Condition	Mean	SD	Spread	n	Failed
economy=AU,terms=english,tokeniser=ascii_only	0.5263	0.0000	0.0000	5	0
economy=AU,terms=english,tokeniser=script_aware	0.5263	0.0000	0.0000	5	0
economy=AU,terms=english_plus_native,tokeniser=ascii_only	0.5263	0.0000	0.0000	5	0
economy=AU,terms=english_plus_native,tokeniser=script_aware	0.5263	0.0000	0.0000	5	0
economy=CN,terms=english,tokeniser=ascii_only	0.0000	0.0000	0.0000	5	0
economy=CN,terms=english,tokeniser=script_aware	0.0000	0.0000	0.0000	5	0
economy=CN,terms=english_plus_native,tokeniser=ascii_only	0.0000	0.0000	0.0000	5	0
economy=CN,terms=english_plus_native,tokeniser=script_aware	0.3947	0.0000	0.0000	5	0
economy=IN,terms=english,tokeniser=ascii_only	0.7959	0.0000	0.0000	5	0
economy=IN,terms=english,tokeniser=script_aware	0.7959	0.0000	0.0000	5	0
economy=IN,terms=english_plus_native,tokeniser=ascii_only	0.7959	0.0000	0.0000	5	0
economy=IN,terms=english_plus_native,tokeniser=script_aware	0.7959	0.0000	0.0000	5	0
economy=MN,terms=english,tokeniser=ascii_only	0.0833	0.0000	0.0000	5	0
economy=MN,terms=english,tokeniser=script_aware	0.0833	0.0000	0.0000	5	0
economy=MN,terms=english_plus_native,tokeniser=ascii_only	0.0833	0.0000	0.0000	5	0
economy=MN,terms=english_plus_native,tokeniser=script_aware	0.2500	0.0000	0.0000	5	0
economy=MY,terms=english,tokeniser=ascii_only	0.8000	0.0000	0.0000	5	0
economy=MY,terms=english,tokeniser=script_aware	0.8000	0.0000	0.0000	5	0
economy=MY,terms=english_plus_native,tokeniser=ascii_only	0.8000	0.0000	0.0000	5	0
economy=MY,terms=english_plus_native,tokeniser=script_aware	0.8000	0.0000	0.0000	5	0
economy=SG,terms=english,tokeniser=ascii_only	0.6364	0.0000	0.0000	5	0
economy=SG,terms=english,tokeniser=script_aware	0.6364	0.0000	0.0000	5	0
economy=SG,terms=english_plus_native,tokeniser=ascii_only	0.6364	0.0000	0.0000	5	0
economy=SG,terms=english_plus_native,tokeniser=script_aware	0.6364	0.0000	0.0000	5	0

C.2 RECALL@10

Condition	Mean	SD	Spread	n	Failed
economy=AU,terms=english,tokeniser=ascii_only	0.2632	0.0000	0.0000	5	0
economy=AU,terms=english,tokeniser=script_aware	0.1579	0.0000	0.0000	5	0
economy=AU,terms=english_plus_native,tokeniser=ascii_only	0.2632	0.0000	0.0000	5	0
economy=AU,terms=english_plus_native,tokeniser=script_aware	0.1579	0.0000	0.0000	5	0
economy=CN,terms=english,tokeniser=ascii_only	0.0000	0.0000	0.0000	5	0
economy=CN,terms=english,tokeniser=script_aware	0.0000	0.0000	0.0000	5	0
economy=CN,terms=english_plus_native,tokeniser=ascii_only	0.0000	0.0000	0.0000	5	0
economy=CN,terms=english_plus_native,tokeniser=script_aware	0.2105	0.0000	0.0000	5	0
economy=IN,terms=english,tokeniser=ascii_only	0.2041	0.0000	0.0000	5	0
economy=IN,terms=english,tokeniser=script_aware	0.2245	0.0000	0.0000	5	0
economy=IN,terms=english_plus_native,tokeniser=ascii_only	0.2041	0.0000	0.0000	5	0
economy=IN,terms=english_plus_native,tokeniser=script_aware	0.2245	0.0000	0.0000	5	0
economy=MN,terms=english,tokeniser=ascii_only	0.0833	0.0000	0.0000	5	0
economy=MN,terms=english,tokeniser=script_aware	0.0833	0.0000	0.0000	5	0
economy=MN,terms=english_plus_native,tokeniser=ascii_only	0.0833	0.0000	0.0000	5	0
economy=MN,terms=english_plus_native,tokeniser=script_aware	0.0833	0.0000	0.0000	5	0
economy=MY,terms=english,tokeniser=ascii_only	0.3029	0.0256	0.0571	5	0
economy=MY,terms=english,tokeniser=script_aware	0.3429	0.0000	0.0000	5	0
economy=MY,terms=english_plus_native,tokeniser=ascii_only	0.3029	0.0256	0.0571	5	0
economy=MY,terms=english_plus_native,tokeniser=script_aware	0.3429	0.0000	0.0000	5	0
economy=SG,terms=english,tokeniser=ascii_only	0.3636	0.0000	0.0000	5	0
economy=SG,terms=english,tokeniser=script_aware	0.3636	0.0000	0.0000	5	0
economy=SG,terms=english_plus_native,tokeniser=ascii_only	0.3636	0.0000	0.0000	5	0
economy=SG,terms=english_plus_native,tokeniser=script_aware	0.3636	0.0000	0.0000	5	0

C.3 MEAN RECIPROCAL RANK

Condition	Mean	SD	Spread	n	Failed
economy=AU,terms=english,tokeniser=ascii_only	0.1450	0.0150	0.0289	5	0
economy=AU,terms=english,tokeniser=script_aware	0.0613	0.0005	0.0013	5	0
economy=AU,terms=english_plus_native,tokeniser=ascii_only	0.1450	0.0150	0.0289	5	0
economy=AU,terms=english_plus_native,tokeniser=script_aware	0.0613	0.0005	0.0013	5	0
economy=CN,terms=english,tokeniser=ascii_only	0.0000	0.0000	0.0000	5	0
economy=CN,terms=english,tokeniser=script_aware	0.0000	0.0000	0.0000	5	0
economy=CN,terms=english_plus_native,tokeniser=ascii_only	0.0000	0.0000	0.0000	5	0
economy=CN,terms=english_plus_native,tokeniser=script_aware	0.1088	0.0002	0.0005	5	0
economy=IN,terms=english,tokeniser=ascii_only	0.0745	0.0010	0.0022	5	0
economy=IN,terms=english,tokeniser=script_aware	0.0807	0.0001	0.0001	5	0
economy=IN,terms=english_plus_native,tokeniser=ascii_only	0.0745	0.0010	0.0022	5	0
economy=IN,terms=english_plus_native,tokeniser=script_aware	0.0807	0.0001	0.0001	5	0
economy=MN,terms=english,tokeniser=ascii_only	0.0833	0.0000	0.0000	5	0
economy=MN,terms=english,tokeniser=script_aware	0.0833	0.0000	0.0000	5	0
economy=MN,terms=english_plus_native,tokeniser=ascii_only	0.0833	0.0000	0.0000	5	0
economy=MN,terms=english_plus_native,tokeniser=script_aware	0.0780	0.0188	0.0424	5	0
economy=MY,terms=english,tokeniser=ascii_only	0.0682	0.0012	0.0029	5	0
economy=MY,terms=english,tokeniser=script_aware	0.0924	0.0001	0.0003	5	0
economy=MY,terms=english_plus_native,tokeniser=ascii_only	0.0682	0.0012	0.0029	5	0
economy=MY,terms=english_plus_native,tokeniser=script_aware	0.0924	0.0001	0.0003	5	0
economy=SG,terms=english,tokeniser=ascii_only	0.1862	0.0028	0.0065	5	0
economy=SG,terms=english,tokeniser=script_aware	0.1780	0.0000	0.0001	5	0
economy=SG,terms=english_plus_native,tokeniser=ascii_only	0.1862	0.0028	0.0065	5	0
economy=SG,terms=english_plus_native,tokeniser=script_aware	0.1780	0.0000	0.0001	5	0